

EXPERIMENT-1-CEASER CIPHER

AIM: To write a C program to implement the Caesar Cipher encryption technique by replacing each letter with another letter positioned k places ahead in the alphabet, where k ranges from 1 to 25.

CODE:

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>

int main()
{
    char str[100];
    int k, i;

    printf("Enter a message: ");
    fgets(str, sizeof(str), stdin);

    printf("Enter key (1-25): ");
    scanf("%d", &k);

    if(k < 1 || k > 25)
    {
        printf("Invalid key!");
        return 0;
    }
```

```
}

for(i = 0; str[i] != '\0'; i++)
{
    if(isupper(str[i]))
        str[i] = (str[i] - 'A' + k) % 26 + 'A';
    else if(islower(str[i]))
        str[i] = (str[i] - 'a' + k) % 26 + 'a';
}

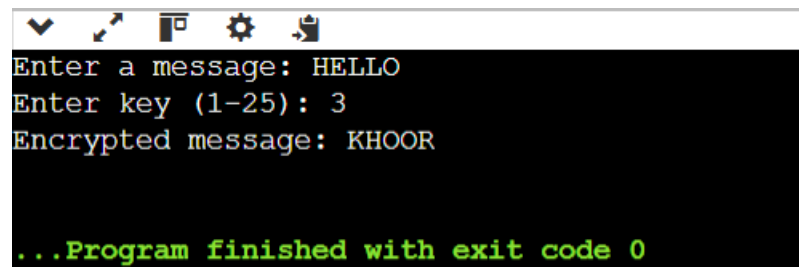
printf("Encrypted message: %s", str);

return 0;
}
```

INPUT:HELLO

KET: 3

OUTPUT: KHOOR



```
Enter a message: HELLO
Enter key (1-25): 3
Encrypted message: KHOOR

...Program finished with exit code 0
```